

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE **COURSE CODE:** MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

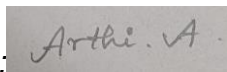
Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I ARTHI A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

AI-Based Autonomous Driving Using the Waymo Driver (Waymo, Alphabet Inc.)

1. Domain Selection

Autonomous Vehicles

2. Introduction and Background

The Waymo Driver is a fully autonomous driving system developed by Waymo (an Alphabet subsidiary), currently operating driverless ride-hailing services in cities such as Phoenix, San Francisco, and Los Angeles. It combines LiDAR, radar, and camera sensors with deep learning-based perception, prediction, and planning models to drive without a human behind the wheel. Unlike driver-assistance systems that support a human driver, the Waymo Driver is designed for full autonomy (SAE Level 4), making it one of the most advanced real-world deployments of AI in transportation.

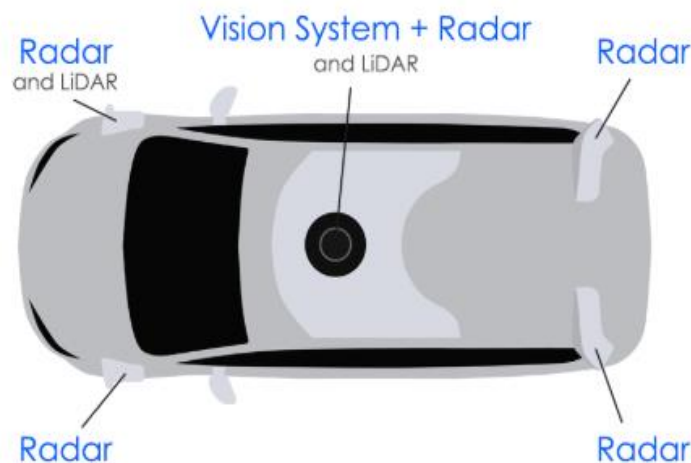


Figure 1: Waymo Driver vehicle showing exterior sensor suite (LiDAR, camera, radar pods)

3. Problem Identification

Human driving error is responsible for the vast majority of road traffic accidents worldwide, caused by factors such as distraction, fatigue, impaired judgment, and slow reaction time. Traditional driver-assistance technologies (cruise control, lane-keeping) reduce some risk but still require constant human attention and intervention. Public transportation and ride-hailing also depend entirely on the availability of human drivers, limiting scalability and consistency of service, particularly in areas or hours with driver shortages.

4. Need for AI Solution

Driving involves interpreting a constantly changing, unpredictable environment — pedestrians, cyclists, other vehicles, weather, and road conditions — in real time. Rule-based programming cannot

account for the near-infinite variety of real-world driving scenarios. AI, particularly deep learning-based perception and prediction models, can learn from millions of miles of driving data to recognize objects, predict their behavior, and make split-second decisions far more consistently than rule-based systems, and without the fatigue or distraction risks inherent to human drivers.

5. Scenario

Consider a Waymo robotaxi navigating a city intersection. Cameras, LiDAR, and radar simultaneously detect a pedestrian stepping toward the crosswalk, a cyclist approaching from the side, and a car edging out from a side street. The AI system must classify each object, predict their likely next movements, and plan a safe path — all within milliseconds — before deciding whether to slow down, stop, or proceed.

6. AI Technologies Used

- **Computer Vision (CNNs):** Detects and classifies objects — pedestrians, vehicles, cyclists, traffic signs — from camera feeds.
- **LiDAR-Based 3D Perception:** Deep learning models process point-cloud data from LiDAR sensors to build a precise 3D map of surrounding objects and distances.
- **Sensor Fusion:** Combines camera, LiDAR, and radar data into a single, unified understanding of the environment, since each sensor type has different strengths.
- **Behavior Prediction Models:** Deep learning models forecast the likely future movement of pedestrians, cyclists, and other vehicles.
- **Path Planning & Reinforcement Learning:** Determines the vehicle's optimal, safe trajectory in real time.

7a. System Architecture — What Are the Components

1. **Sensor Suite** – Cameras, LiDAR units, and radar sensors mounted around the vehicle
2. **Sensor Fusion Module** – Combines all sensor inputs into one unified environmental model
3. **Perception Module** – CNN-based detection to identify pedestrians, vehicles, cyclists, signs
4. **Prediction Module** – Forecasts the likely future movement of detected objects
5. **Planning/Decision-Making Module** – Computes the vehicle's safe trajectory
6. **Vehicle Control** – Executes steering, braking, and acceleration

7b. Why These Components (Justification)

- **Sensor suite:** no single sensor works reliably in every condition — cameras struggle in poor light, LiDAR in heavy rain, radar lacks fine detail.

- **Sensor fusion:** merges these inputs into one coherent, high-confidence view, critical for safety-critical decisions.
- **Perception:** converts raw sensor data into identified objects — without this, the vehicle has no understanding of its surroundings.
- **Prediction:** necessary to anticipate what pedestrians and vehicles will do next, not just where they currently are.
- **Planning:** the final step that turns understanding into an actual driving action.
- **Vehicle control:** executes the decision physically and must respond in real time.

8. Working Mechanism

The Waymo Driver continuously captures data from its camera, LiDAR, and radar sensors. Sensor fusion combines these inputs into a unified 3D representation of the environment. Perception models then classify all detected objects, while prediction models estimate how each object is likely to move in the next few seconds. This information feeds into a planning module that computes the safest trajectory, executed through the car's steering, braking, and acceleration systems — repeated many times per second.

9. Data Sources and Processing

Data sources include real-time sensor streams (camera images, LiDAR point clouds, radar signals) as well as pre-built HD maps of the operating area. Raw sensor data is calibrated and filtered, then passed through perception models trained on millions of miles of real and simulated driving data. Edge cases encountered on the road are logged and used to continuously retrain and improve models across the fleet.

10. Real-World Implementation — Where It Is Used

- Robotaxi services: driverless ride-hailing in Phoenix, San Francisco, and Los Angeles
- Airport transportation: driverless service to airport routes in select cities
- Urban commuting: fully driverless trips via the Waymo One app

11. Why It Is Used There

Dense urban environments with complex traffic patterns provide both the greatest challenge and greatest value for autonomous driving — reducing accidents from human error, providing consistent mobility regardless of driver availability, and offering transportation access to those who cannot drive. Cities with favorable weather and well-mapped roads are typically chosen first, reducing technical difficulty.

12. Impact and Benefits

Autonomous driving systems aim to reduce accidents caused by human error such as distraction or impairment. They provide consistent transportation availability without dependency on human driver schedules, and fleet-wide learning allows continuous improvement based on real-world driving experience across thousands of vehicles.

13. Challenges and Limitations

Autonomous vehicles face challenges handling rare, unpredictable "edge case" scenarios, such as unusual construction or erratic human behavior. Adverse weather like heavy snow or fog can degrade sensor performance. Regulatory approval varies by region, public trust remains an ongoing challenge, and the cost of sensor hardware (particularly LiDAR) remains a barrier to wider adoption.

14. Future Scope and Emerging Trends

Future development includes expanding operation to highway driving and wider weather conditions, further cost reduction of LiDAR hardware, and improved simulation for rare edge cases. There is growing interest in vehicle-to-infrastructure (V2I) communication and road infrastructure.

15. Conclusion

The Waymo Driver demonstrates a real-world, commercially deployed example of how AI-based perception, prediction, and planning enable fully autonomous driving in complex urban environments. By fusing multiple sensor types and continuously learning from fleet-wide data, the system improves road safety and transportation accessibility, while challenges around edge cases, weather, and regulation remain areas of ongoing progress.

References

1. Waymo LLC. *Waymo Driver — Technology Overview*. Official Product Documentation.
2. Bojarski, M., et al. (2016). End to End Learning for Self-Driving Cars. *arXiv preprint*.
3. Levinson, J., et al. (2011). Towards fully autonomous driving: Systems and algorithms. *IEEE Intelligent Vehicles Symposium*.
4. Badue, C., et al. (2021). Self-driving cars: A survey. *Expert Systems with Applications*.